Math 253: Quiz 5 **15 October, 2026**

Name: _____ **/ 25**

30 minutes maximum. No aids (book, calculator, etc.) are permitted. Show all work and use proper notation for full credit. Answers should be in reasonably-simplified form. 25 points possible.

1. Consider the function $f(x,y) = x^2 + xy + y^2 - 3x$.

   (a) (4 points) Find all critical points of f.

   (b) (3 points) Classify the critical points using $D(x_0,y_0) = f_{xx}(x_0,y_0)f_{yy}(x_0,y_0) - (f_{xy}(x_0,y_0))^2$.

2. (6 points) Find an equation for the tangent plane to the surface $-4x^3 + 12y^2 + 4z = -4$ at the point $(2,1,4)$.

3. Consider the function $f(x,y) = e^x \cos(y)$.

   (a) (6 points) Find the directional derivative of f in the direction of the vector $\langle 3, -4 \rangle$ at the point $(0, \frac{\pi}{2})$.

   (b) (3 points) Find the direction of the maximum rate of change of f at the point $(0, \frac{\pi}{2})$.

   (c) (3 points) Find the maximum rate of change of f at the point $(0, \frac{\pi}{2})$.